

scores to determine if samples are out-of-distribution.

Bayesian neural networks (Denker & Lecun, 1990; MacKay, 1992) return a probability distribution over outputs as an alternative way to represent model uncertainty. Gal & Ghahramani (2016) draw a connection between Dropout (Srivastava et al., 2014) and model uncertainty, claiming that sampling models with dropped nodes is a way to estimate the probability distribution over all possible models for a given sample. Kendall & Gal (2017) combine this approach with a model that outputs a predictive mean and variance for each data point. This notion of uncertainty is not restricted to classification problems. Additionally, neural networks can be used in conjunction with Bayesian models that output complete distributions. For example, deep kernel learning (Wilson et al., 2016a;b; Al-Shedivat et al., 2016) combines deep neural networks with Gaussian processes on classification and regression problems. In contrast, our framework, which does not augment the neural network model, returns a confidence score rather than returning a distribution of possible outputs.

5. Results

We apply the calibration methods in Section 4 to image classification and document classification neural networks. For image classification we use 6 datasets:

1. Caltech-UCSD Birds (Welinder et al., 2010): 200 bird species. 5994/2897/2897 images for train/validation/test sets.
2. Stanford Cars (Krause et al., 2013): 196 classes of cars by make, model, and year. 8041/4020/4020 images for train/validation/test.
3. ImageNet 2012 (Deng et al., 2009): Natural scene images from 1000 classes. 1.3 million/25,000/25,000 images for train/validation/test.
4. CIFAR-10/CIFAR-100 (Krizhevsky & Hinton, 2009): Color images (32×32) from 10/100 classes. 45,000/5,000/10,000 images for train/validation/test.
5. Street View House Numbers (SVHN) (Netzer et al., 2011): 32×32 colored images of cropped out house numbers from Google Street View. 598,388/6,000/26,032 images for train/validation/test.

We train state-of-the-art convolutional networks: ResNets (He et al., 2016), ResNets with stochastic depth (SD) (Huang et al., 2016), Wide ResNets (Zagoruyko & Komodakis, 2016), and DenseNets (Huang et al., 2017). We use the data preprocessing, training procedures, and hyperparameters as described in each paper. For Birds and Cars, we fine-tune networks pretrained on ImageNet.

For document classification we experiment with 4 datasets:

1. 20 News: News articles, partitioned into 20 cate-

gories by content. 9034/2259/7528 documents for train/validation/test.

2. Reuters: News articles, partitioned into 8 categories by topic. 4388/1097/2189 documents for train/validation/test.
3. Stanford Sentiment Treebank (SST) (Socher et al., 2013): Movie reviews, represented as sentence parse trees that are annotated by sentiment. Each sample includes a coarse binary label and a fine grained 5-class label. As described in (Tai et al., 2015), the training/validation/test sets contain 6920/872/1821 documents for binary, and 544/1101/2210 for fine-grained.

On 20 News and Reuters, we train Deep Averaging Networks (DANs) (Iyyer et al., 2015) with 3 feed-forward layers and Batch Normalization. On SST, we train TreeLSTMs (Long Short Term Memory) (Tai et al., 2015). For both models we use the default hyperparameters suggested by the authors.

Calibration Results. Table 1 displays model calibration, as measured by ECE (with $M = 15$ bins), before and after applying the various methods (see Section S3 for MCE, NLL, and error tables). It is worth noting that most datasets and models experience some degree of miscalibration, with ECE typically between 4 to 10%. This is not architecture specific: we observe miscalibration on convolutional networks (with and without skip connections), recurrent networks, and deep averaging networks. The two notable exceptions are SVHN and Reuters, both of which experience ECE values below 1%. Both of these datasets have very low error (1.98% and 2.97%, respectively); and therefore the ratio of ECE to error is comparable to other datasets.

Our most important discovery is the *surprising effectiveness of temperature scaling* despite its remarkable simplicity. Temperature scaling outperforms all other methods on the vision tasks, and performs comparably to other methods on the NLP datasets. What is perhaps even more surprising is that temperature scaling outperforms the vector and matrix Platt scaling variants, which are strictly more general methods. In fact, vector scaling recovers essentially the same solution as temperature scaling – the learned vector has nearly constant entries, and therefore is no different than a scalar transformation. In other words, network miscalibration is intrinsically low dimensional.

The only dataset that temperature scaling does not calibrate is the Reuters dataset. In this instance, only one of the above methods is able to improve calibration. Because this dataset is well-calibrated to begin with ($ECE \leq 1\%$), there is not much room for improvement with any method, and post-processing may not even be necessary to begin with. It is also possible that our measurements are affected by dataset split or by the particular binning scheme.